

DMV CoreTech Intern Task 1

Python Scientific Calculator

Name: Arya Gupta

Domain: Data Science

GitHub Repository Link: <https://github.com/iaryaguptaa/Python-Scientific-Calculator>

Project Implementation

- **Modular Architecture:** Instead of a single script, I separated each mathematical operation (addition, division, logarithms, etc.) into its own independent module. This mimics the architecture of real-world Data Science libraries like Pandas or NumPy.
- **Centralized Interface:** The main.py file acts solely as the user interface, importing the required modules and handling the user's menu choices.
- **Robust Error Handling:** I implemented try...except ValueError blocks to catch invalid inputs (like typing letters instead of numbers) and added mathematical edge-case handling (e.g., preventing crashes when dividing by zero or taking the log of a negative number).

Output Screenshots

```
Select an operation (1-11): 4
Enter first number: 15
Enter second number: 0
Result: Error: Cannot divide by zero!
```

```
Scientific Calculator
1. Addition (+)
2. Subtraction (-)
3. Multiplication (*)
4. Division (/)
5. Power (x^y)
6. Square Root (√x)
7. Trigonometry (sin, cos, tan)
8. Logarithms (ln, lg)
9. Angle Conversions (Deg<->Rad)
10. Pi (π)
11. Exit
```

```
Select an operation (1-11): █
```

```
Select an operation (1-11): 8
Enter the number: 0
Choose (a) ln, (b) lg: b
Result: Error: Logarithm is undefined for zero or negative numbers.
```

```
Scientific Calculator
1. Addition (+)
2. Subtraction (-)
3. Multiplication (*)
4. Division (/)
5. Power (x^y)
6. Square Root (√x)
7. Trigonometry (sin, cos, tan)
8. Logarithms (ln, lg)
9. Angle Conversions (Deg<->Rad)
10. Pi (π)
11. Exit
```

```
Select an operation (1-11): █
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
~/Users/ARYA/Desktop/DS Internship projects/1) calculator 26 May 2026/main.py"
```

```
Scientific Calculator
1. Addition (+)
2. Subtraction (-)
3. Multiplication (*)
4. Division (/)
5. Power (x^y)
6. Square Root (√x)
7. Trigonometry (sin, cos, tan)
8. Logarithms (ln, lg)
9. Angle Conversions (Deg<->Rad)
10. Pi (π)
11. Exit
```

```
Select an operation (1-11): 1
Enter first number: 5.5
Enter second number: -2.2
Result: 3.3
```

```
Scientific Calculator
1. Addition (+)
2. Subtraction (-)
3. Multiplication (*)
4. Division (/)
5. Power (x^y)
6. Square Root (√x)
7. Trigonometry (sin, cos, tan)
8. Logarithms (ln, lg)
9. Angle Conversions (Deg<->Rad)
10. Pi (π)
11. Exit
```

```
Select an operation (1-11): █
```

1. Basic to Intermediate Level Questions

Q1. What is the difference between a List and a Tuple in Python?

The primary difference is mutability.

- **Lists** are mutable, meaning their elements can be changed, added, or removed after creation. They are defined using square brackets `[]`.
- **Tuples** are immutable, meaning once they are created, their elements cannot be modified. They are defined using parentheses `()`. Because they are immutable, tuples are faster and more memory-efficient than lists.

Q2. Explain the difference between mutable and immutable data types. Give examples.

- **Mutable** data types allow their state or contents to be changed in-place without creating a new object in memory. *Examples:* Lists, Dictionaries, Sets.
- **Immutable** data types do not allow any changes after they are created. Any operation that attempts to modify them will instead create a brand new object in memory. *Examples:* Strings, Integers, Floats, Tuples.

Q3. What is a Dictionary in Python and how is it structured?

A dictionary is an unordered, mutable collection of data values stored as **key-value pairs**. It is defined using curly braces `{}`. Each key in a dictionary must be unique and of an immutable data type (like a string or integer), while the values can be of any data type and can be duplicated. They are heavily used in Data Science for JSON processing and data mapping.

2. Concept-Based Questions

Q4. What is the difference between 'break', 'continue', and 'pass'?

- **break** : Terminates the loop entirely and transfers execution to the statement immediately following the loop.
- **continue** : Skips the remaining code inside the loop for the current iteration and forces the loop to move to the next iteration.
- **pass** : A null operation. It does nothing when executed. It is used as a placeholder in empty functions or loops to prevent syntax errors.

Q5. How does Python manage memory?

Python manages memory automatically using two primary mechanisms:

1. **Reference Counting**: Python keeps track of the number of references to every object in memory. If an object's reference count drops to zero, the memory is immediately freed.
2. **Garbage Collection**: A background process that specifically looks for and cleans up "reference cycles" (e.g., when object A references object B, and object B references object A, preventing their reference counts from ever reaching zero).

Q6. What is the purpose of the `__init__` method in object-oriented programming?

The `__init__` method is a special method (also known as a constructor) in Python classes. It is automatically called when a new instance (object) of a class is created. Its primary purpose is to initialize the attributes of the object, assigning initial values to the object's properties.

3. Output-Based Questions

Q7. What will be the output of the following code? Explain why.

```
list_a = [1, 2, 3]
list_b = list_a
list_b.append(4)
print("list_a:", list_a)
print("list_b:", list_b)
```

Output:

```
list_a: [1, 2, 3, 4]
```

```
list_b: [1, 2, 3, 4]
```

Explanation: In Python, assigning an existing list to a new variable (`list_b = list_a`) does not create a copy of the list. Instead, both variables point to the exact same object in memory. Therefore, modifying `list_b` also modifies `list_a` .

Q8. What will be the output of the following string slicing operation?

```
word = "DataScience"
print(word[::-1])
```

Output:

```
ecneicSataD
```

Explanation: The slicing syntax is `[start:stop:step]` . Leaving start and stop blank tells Python to cover the entire string, and a step of `-1` tells it to traverse the string backwards, effectively reversing it.

Q9. Analyze the dictionary output.

```
my_dict = {"a": 1, "b": 2, "a": 3}
print(my_dict)
```

Output:

```
{'a': 3, 'b': 2}
```

Explanation: Dictionary keys must be unique. If a key is repeated during dictionary creation, Python will overwrite the previous value with the last value assigned to that key.

4. Short Coding Questions

Q10. Write a list comprehension to create a list of the squares of all even numbers from 1 to 10.

```
squares_of_evens = [x**2 for x in range(1, 11) if x % 2 == 0]
print(squares_of_evens)
# Output: [4, 16, 36, 64, 100]
```

Q11. Write a short function using Exception Handling to safely divide two numbers.

```
def safe_divide(a, b):
    try:
        result = a / b
        return result
    except ZeroDivisionError:
        return "Error: Cannot divide by zero!"
    except TypeError:
        return "Error: Both inputs must be numbers."

print(safe_divide(10, 0))
```

Q12. Write a short Python program to check if a string is a palindrome.

```
def is_palindrome(text):  
    # Convert to lowercase to ensure case-insensitivity  
    text = text.lower()  
    return text == text[::-1]  
  
print(is_palindrome("Radar")) # Output: True
```

5. Frequently Asked Interview Questions

Q13. What is the difference between the '==' operator and the 'is' keyword?

- **==** (Equality Operator): Checks if the **values** of two operands are equal.
- **is** (Identity Operator): Checks if two operands refer to the exact same **object in memory**.

Example: Two identical lists will return True for **==**, but False for **is** because they exist at different memory addresses.

Q14. What are *args and **kwargs in Python functions?

They are used to pass a variable number of arguments to a function.

- ***args** (Non-Keyword Arguments): Allows passing multiple positional arguments. Inside the function, they are gathered into a **tuple**.
- ****kwargs** (Keyword Arguments): Allows passing multiple named (keyword) arguments. Inside the function, they are gathered into a **dictionary**.

Q15. Explain the concept of 'self' in Python object-oriented programming.

self represents the instance of the class itself. When defining methods inside a class, **self** must be the first parameter. It binds the attributes and methods to the specific object being created, allowing the object to access its own data. It is similar to **this** in Java or C++.

Q16. What is the safest way to open and read a file in Python?

The safest and most Pythonic way to handle files is using the `with` statement (a context manager). It ensures that the file is automatically closed as soon as the block of code exits, even if an exception occurs during file processing.

```
with open("data.txt", "r") as file:
    content = file.read()
    print(content)
# File is automatically closed here
```


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SUMMARY

A SECOND-YEAR B.TECH STUDENT IN AI AND DATA SCIENCE. I USE MICROSOFT EXCEL EFFECTIVELY FOR TRACKING AND REPORTING DATA. RIGHT NOW, I'M TAKING STRUCTURED CLASSES TO IMPROVE MY PYTHON AND DATA ANALYSIS SKILLS. I AM A QUICK LEARNER WHO GETS THINGS DONE EVEN UNDER PRESSURE AND TIGHT DEADLINES. PROVEN BY SECURING 2ND POSITION AT ROBOLEAGUE '26.

EDUCATION

Arya College of Engineering & IT

B.Tech in AI & Data Science (2nd Year)
2024 - Present

St. Teresa School

Class 12 (CBSE)

85.2%

Class 10 (CBSE)

89.2%

TECHNICAL SKILLS

- **Microsoft Excel (Intermediate)**
- **Python (Beginner - Actively Learning)**
- **Tools:** Git and GitHub (for version control and managing repositories), Google Colab, VS Code, and Microsoft Excel
- **AI Tools:** ChatGPT for productivity and problem-solving assistance
- **Certifications Earned:** IIT Bombay Spoken Tutorial (C - 82.5%), Cisco Networking Academy (C Essentials I), Infosys Springboard (C), and Simplilearn (Git & Version Control).

PROJECTS

ExcelPersonalExpenseTracker [GitHub • Microsoft Excel]

- Used Ms Excel to make a simple expense tracker with a summary and chart view for each category.

Simple-Calculator-in-C [GitHub • C]

- Made a simple calculator program in C that can do basic math.

Python To-Do List App [GitHub • Google Colab]

- Used Google Colab to make a simple Python To-Do List App.

Portfolio Website [GitHub • HTML]

- I made my own portfolio website.

COURSES

CodeWithHarry - Data Analytics Course (In Progress):

A full course for data analysts that covers Excel, Python, SQL, Power BI, Tableau, and AI tools

ACHIEVEMENTS

Roboleague '26 - 2nd Position:

Won 2nd place in the "VAYUSHTRA" technical competition at Arya College of Engineering & IT in February 2026. Showed that they could work together and solve problems under pressure in a competition.

Hackstorm (24-Hour Hackathon):

Took part in a hackathon put on by JEC Kukas and Hacker's Unity in November 2025.

Freshers Event - 2nd RunnerUp

Fashion Show - 2nd Place

LEADERSHIP & CO-CURRICULAR

Dance Club Coordinator

Managing team schedules, coordinating performances, and organising cultural events at Arya College shows that you can lead, communicate, and plan.

Cultural Participation:

Actively participating in college fests including 'Victory' and inter-college events

ARYA GUPTA

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ABOUT ME

2nd year B.Tech student in AI & Data Science with hands-on experience in Web Development, UI/UX Design and Data Analysis using Excel and Python. Well versed in developing real time projects on own and delivering within deadlines. Actively upskilling in Full Stack development with a focus on HTML, CSS, Javascript & React.

EDUCATION

Arya College of Engineering & IT

B.Tech in AI & Data Science (2nd Year) | 2024 – Present
CGPA: 8.03 (Sem 3 SGPA: 8.37)

St. Teresa School, Jaipur

Class 12 (CBSE): 85.2%
Class 10 (CBSE): 89.2%

Relevant Coursework

- C, C++, OOPs, DSA
- Web Dev: HTML, CSS, JS, React
- DBMS, Linux, Software Engineering
- Digital Electronics, Discrete Mathematics

TECHNICAL SKILLS

Languages & Web

- HTML / CSS — Intermediate
- JavaScript — Beginner
- React — Basic
- Python — Beginner (Actively Learning)

Tools & Platforms

- Microsoft Excel — Intermediate
- Git & GitHub — Version Control
- Google Colab, VS Code
- Canva — UI/UX Design (Basic)

AI & Productivity

- ChatGPT for productivity & problem-solving
- AI Tools Workshop (be10x Certified)

Certifications

- IIT Bombay Spoken Tutorial — C Programming (82.5%)
- Cisco Networking Academy — C Essentials 1
- Infosys Springboard — C
- Simplilearn — Git & Version Control
- be10x — AI Tools & ChatGPT Workshop (Apr 2026)

PROJECTS

- Funngro Website Revamp [HTML · CSS · GitHub]
- SkillUp Academy - EdTech UI Design [Canva · UI/UX Design]
- Excel Personal Expense Tracker [Microsoft Excel · GitHub]
- Portfolio Website [HTML · CSS · GitHub]
- Python To-Do List App [Python · Google Colab · GitHub]

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Covers Excel, Python, SQL, Power BI, Tableau, and AI Tools. Currently completing Python and SQL modules.

ACHIEVEMENTS

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Won 2nd place at Arya College's technical competition, Feb 2026. Demonstrated problem-solving and teamwork under pressure.

Hackstorm — 24-Hour Hackathon

Participated in hackathon by JEC Kukas & Hacker's Unity, Nov 2025.

Freshers Event — 2nd Runner Up

Recognized at college fresher's event for overall performance.

Fashion Show — 2nd Place

Secured 2nd place at college cultural fashion show event.

LEADERSHIP & CO-CURRICULAR

Dance Club Coordinator

Managed team schedules, coordinated performances, and organised cultural events at Arya College — demonstrating leadership, communication, and planning skills.

Cultural Participation

Actively participated in college fests including 'Victory' and various inter-college cultural events.

Name: Arya Gupta

Project: Funngro Website Revamp

Live Website Link: <https://iaryaguptaa.github.io/funngro-website-revamp/#>

GitHub Repository Link: <https://github.com/iaryaguptaa/funngro-website-revamp>





